

FDT PE Factory Data Study

Dr. Scientist

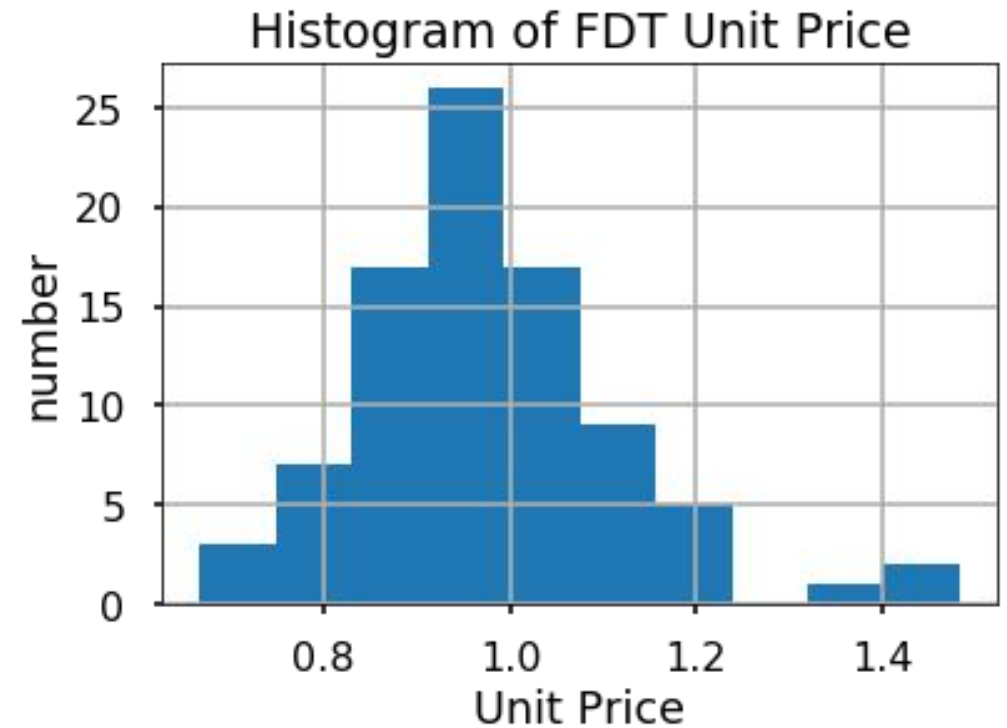
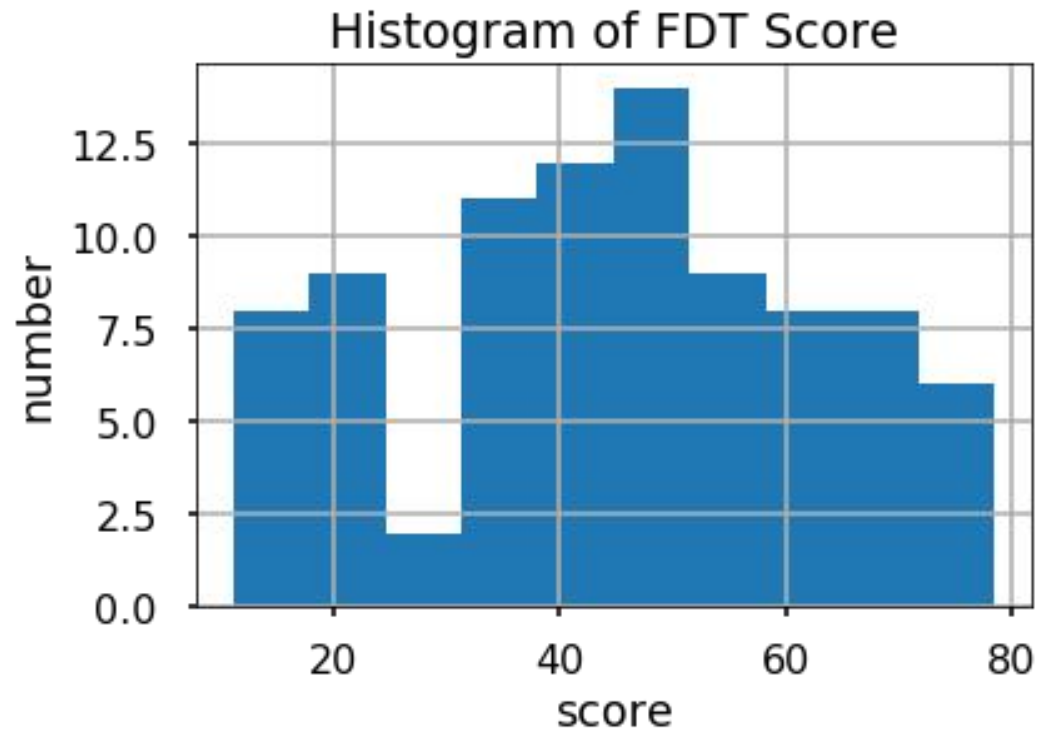
2017-Nov-22

Outline



- **PE Factory with FDTScore**
- Portfolio Strategies Details

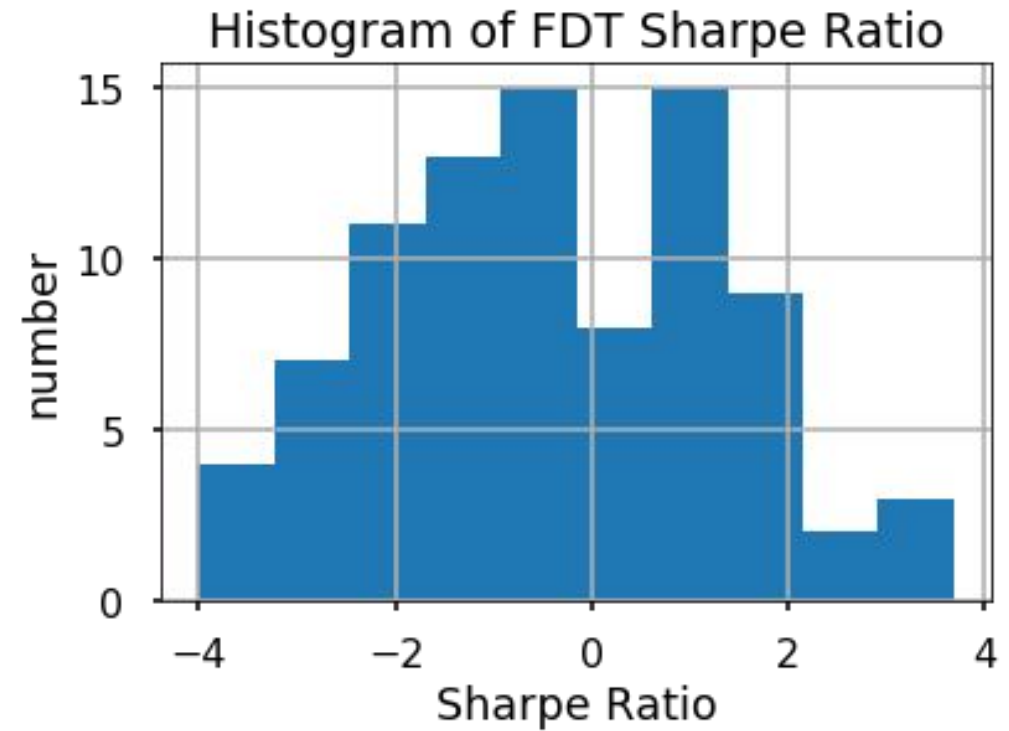
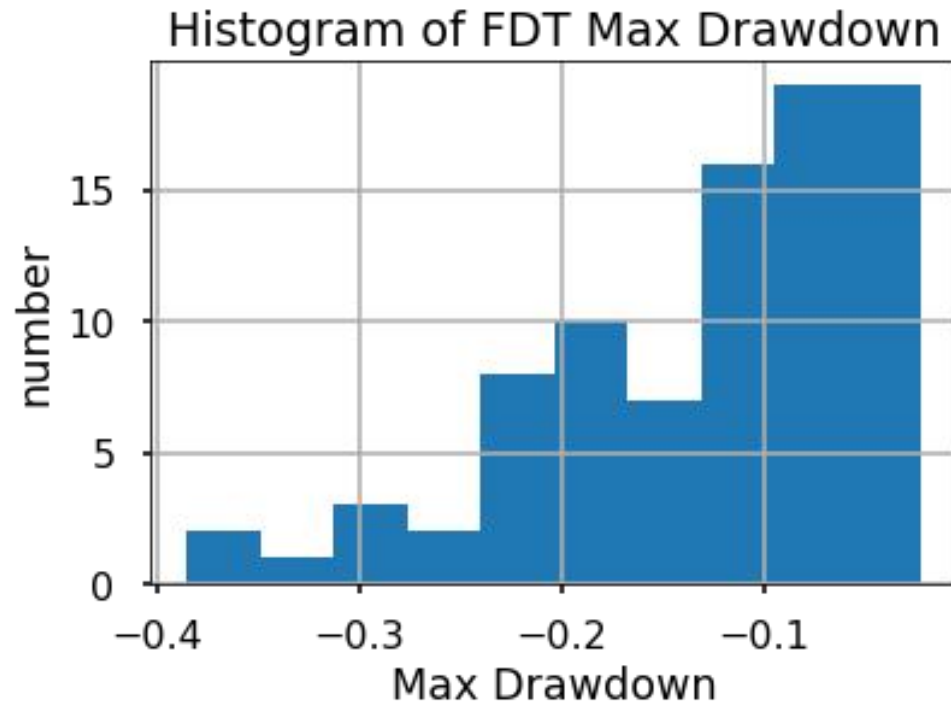
FDTScore Statistics



Overall PnL of all traders up to 2017-11-21

- USD: -107,127.00

FDTScore Statistics

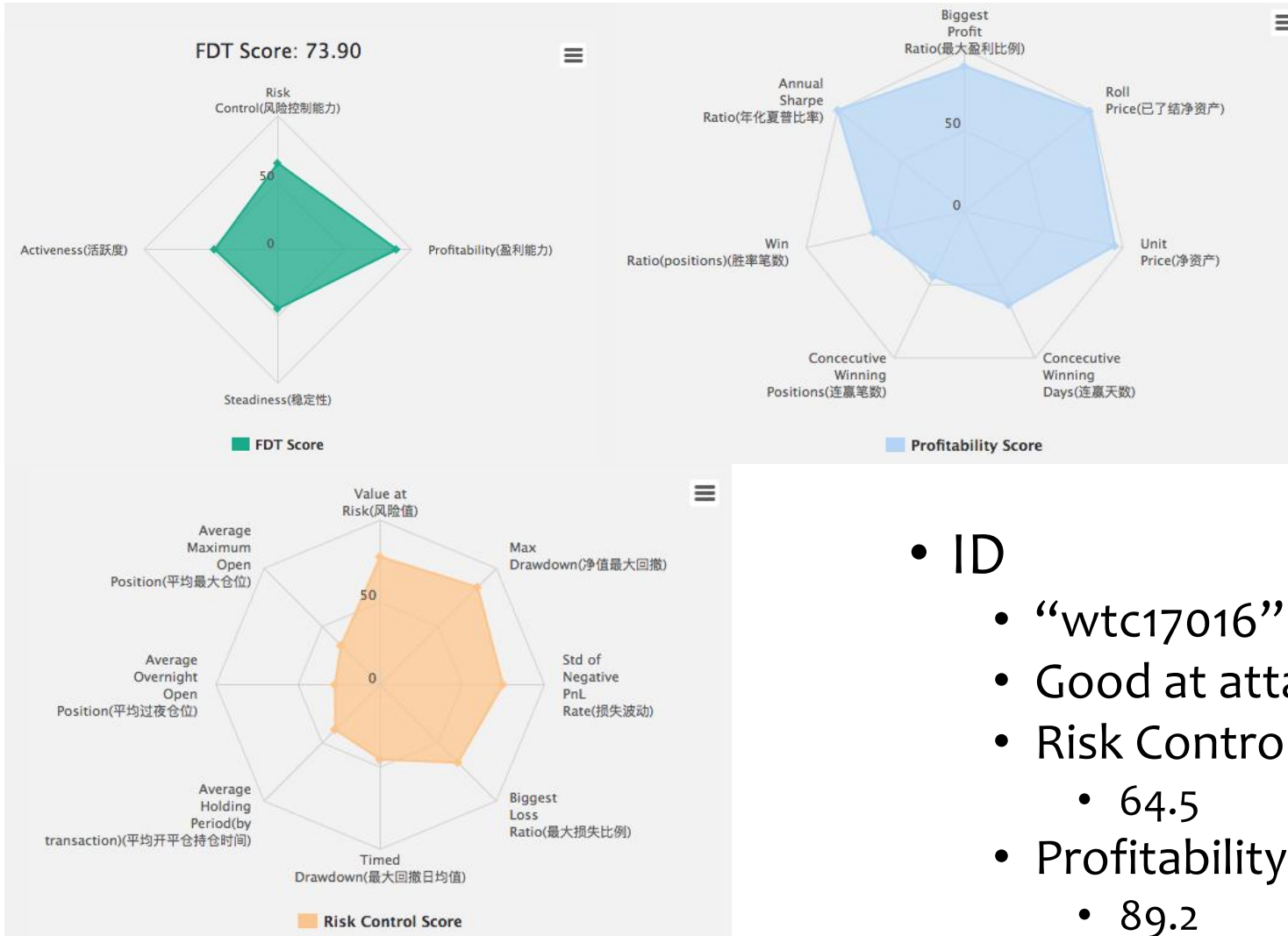


Talents thru FDTScore



- ID
 - “wtc17009”
 - Balanced skills
 - Risk Control
 - 80.6
 - Profitability
 - 80.5

Talents thru FDTScore



- ID
 - “wtc17016”
 - Good at attacking
 - Risk Control
 - 64.5
 - Profitability
 - 89.2

Outline

- PE Factory with FDTScore
- **Detail about Portfolio Strategies**
 - **Study Items**
 - **FDTScore: Diversity Analysis**
 - **FDTScore: Soft-unique Analysis**
 - **FDTScore: Hedging Effect Analysis**

Portfolio Strategy “beecu001”

U.S. Stock

FDTAI



• Portfolio “beecu001”

- Started on 2017-08-14 (68 days)
- Market: US Stock
- Benchmark: S&P 500
- Alpha: -0.02
- Beta: 1.05
- Max Drawdown: -3.3%
- Annual Ret.: 21%
- Annual Volatility: 8.9%

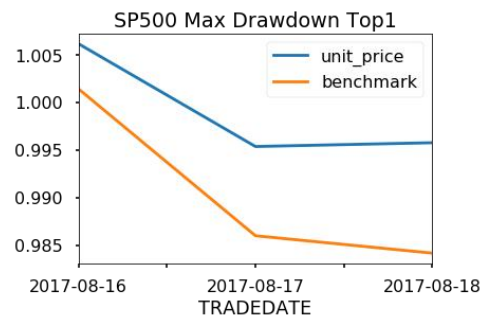
Portfolio Strategy “beecu001”

Blue: strategy

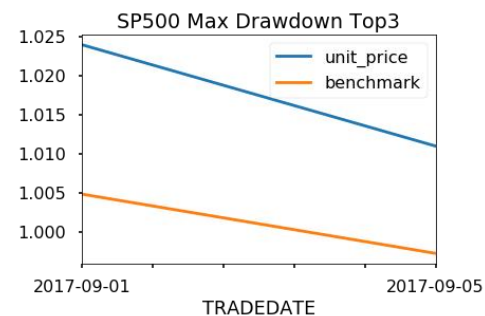
Orange: Benchmark

FDTAI

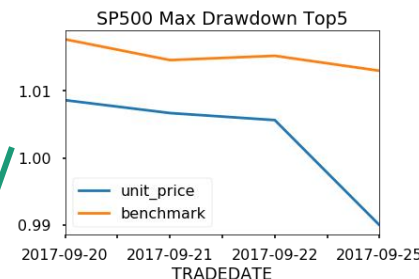
- Drawdown analysis



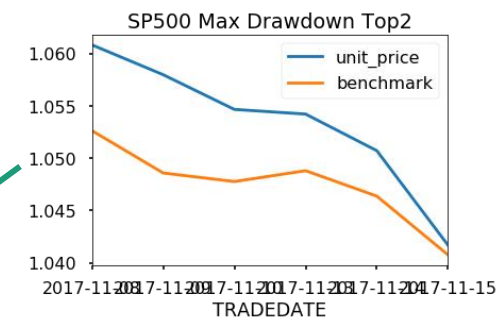
Top-1



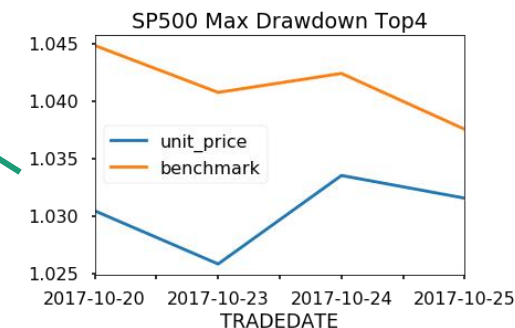
Top-3



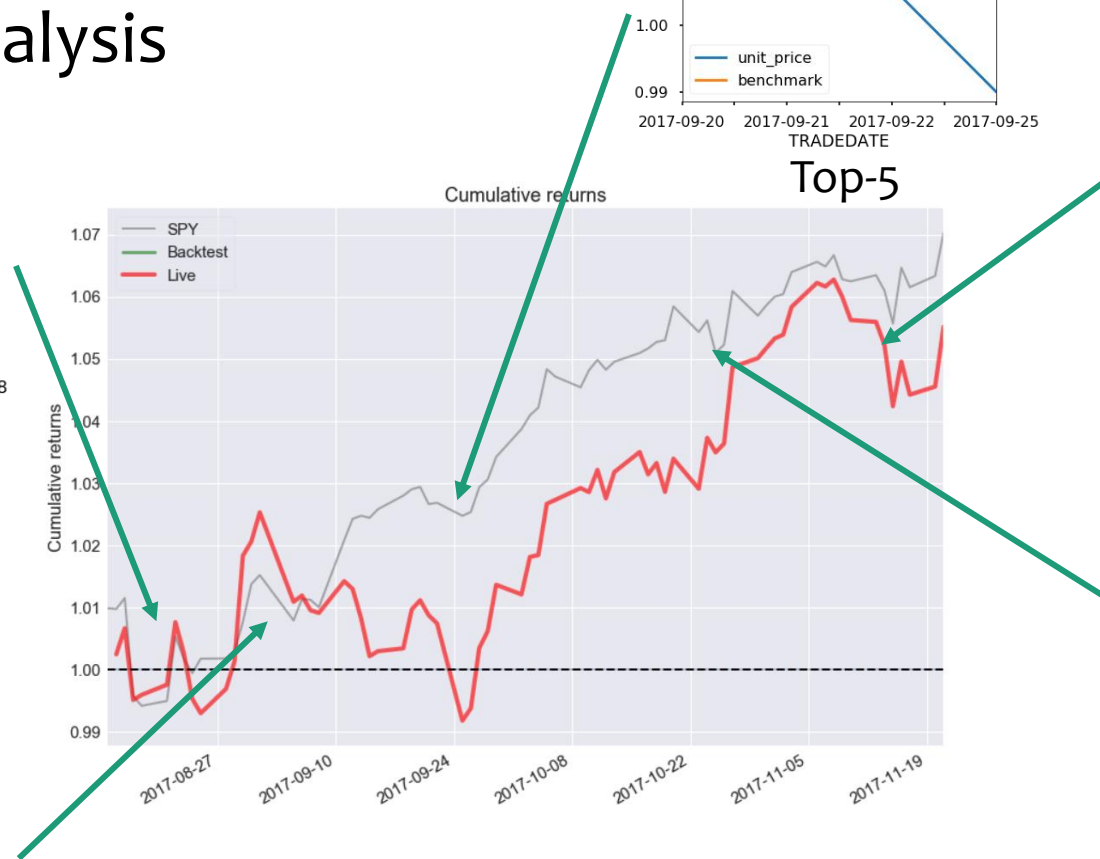
Top-5



Top-2



Top-4



FDTScore: Diversity Analysis - "beecu001"

U.S. Stock

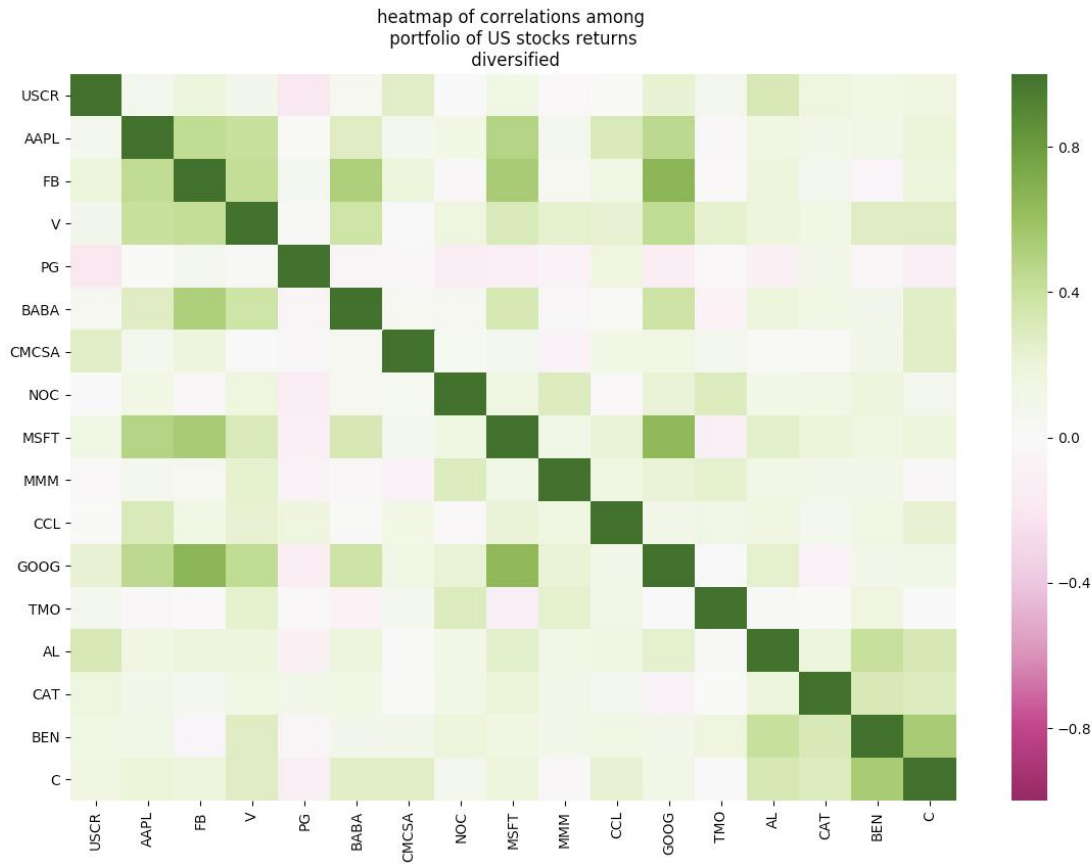
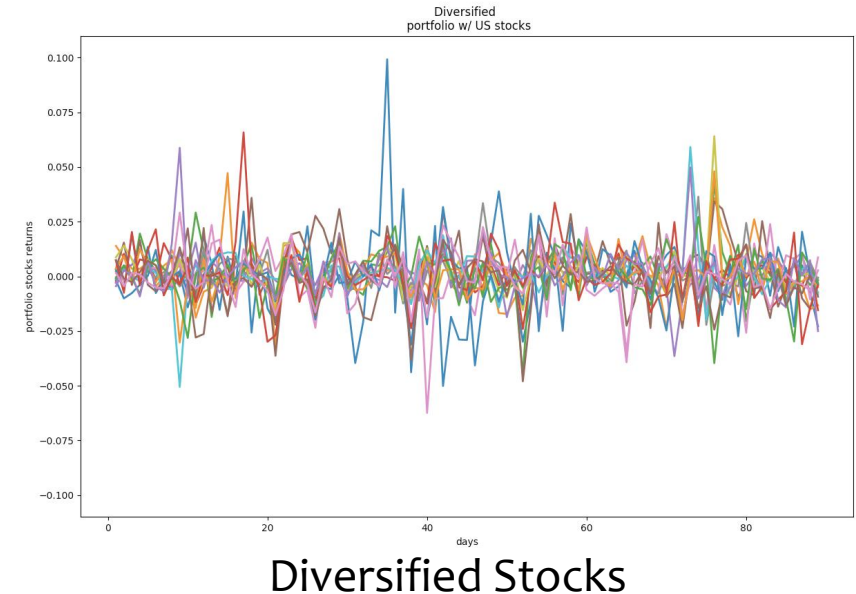


Fig. 1 Heatmap of correlations among portfolio assets' returns - Diversified (beecu001)

Portfolio Stocks Returns



- Remarks
 - Both plots shows the trader's portfolio being well diversified.

FDTScore: Soft-unique Analysis – ‘beecu001’^{U.S. Stock}



Fig. 3 Number of assets explaining portfolio variance

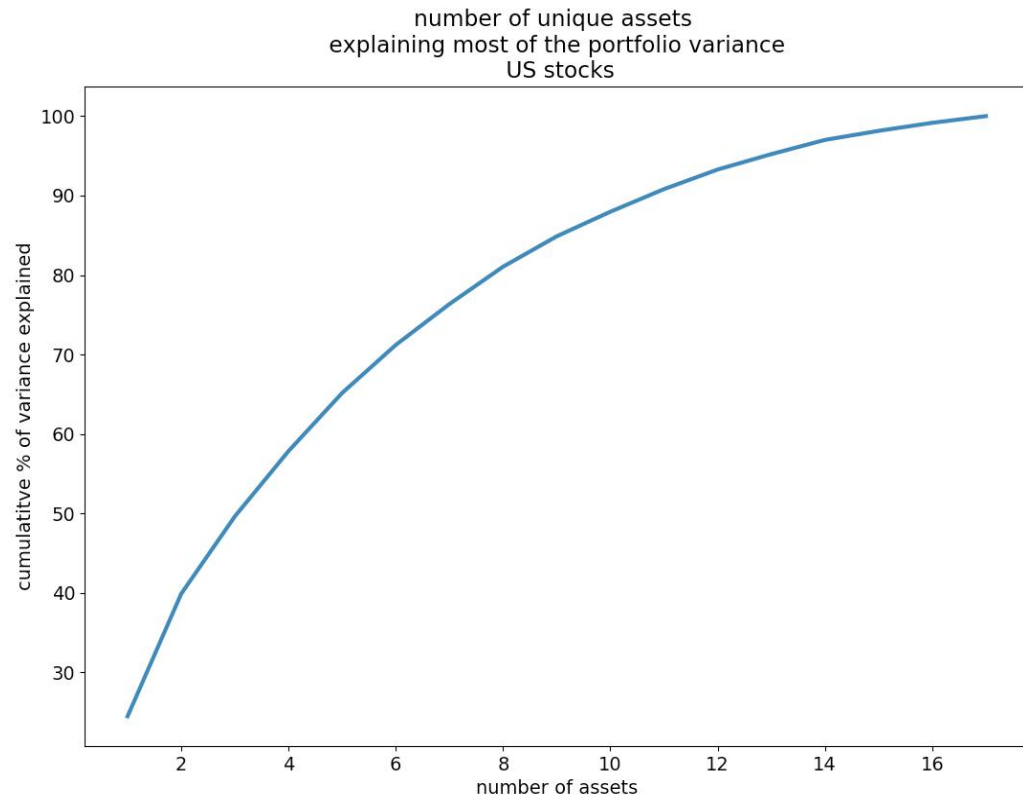
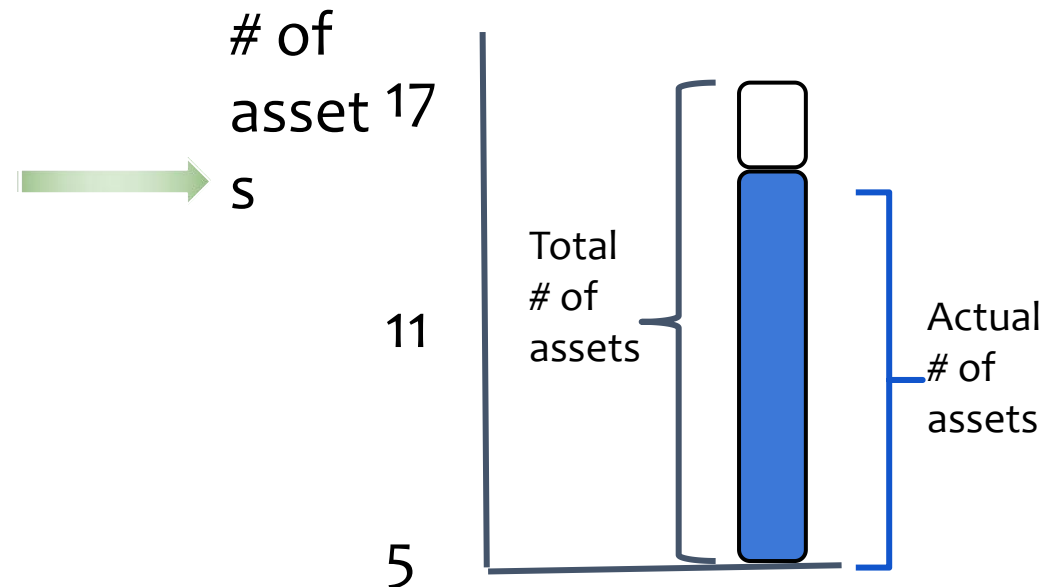
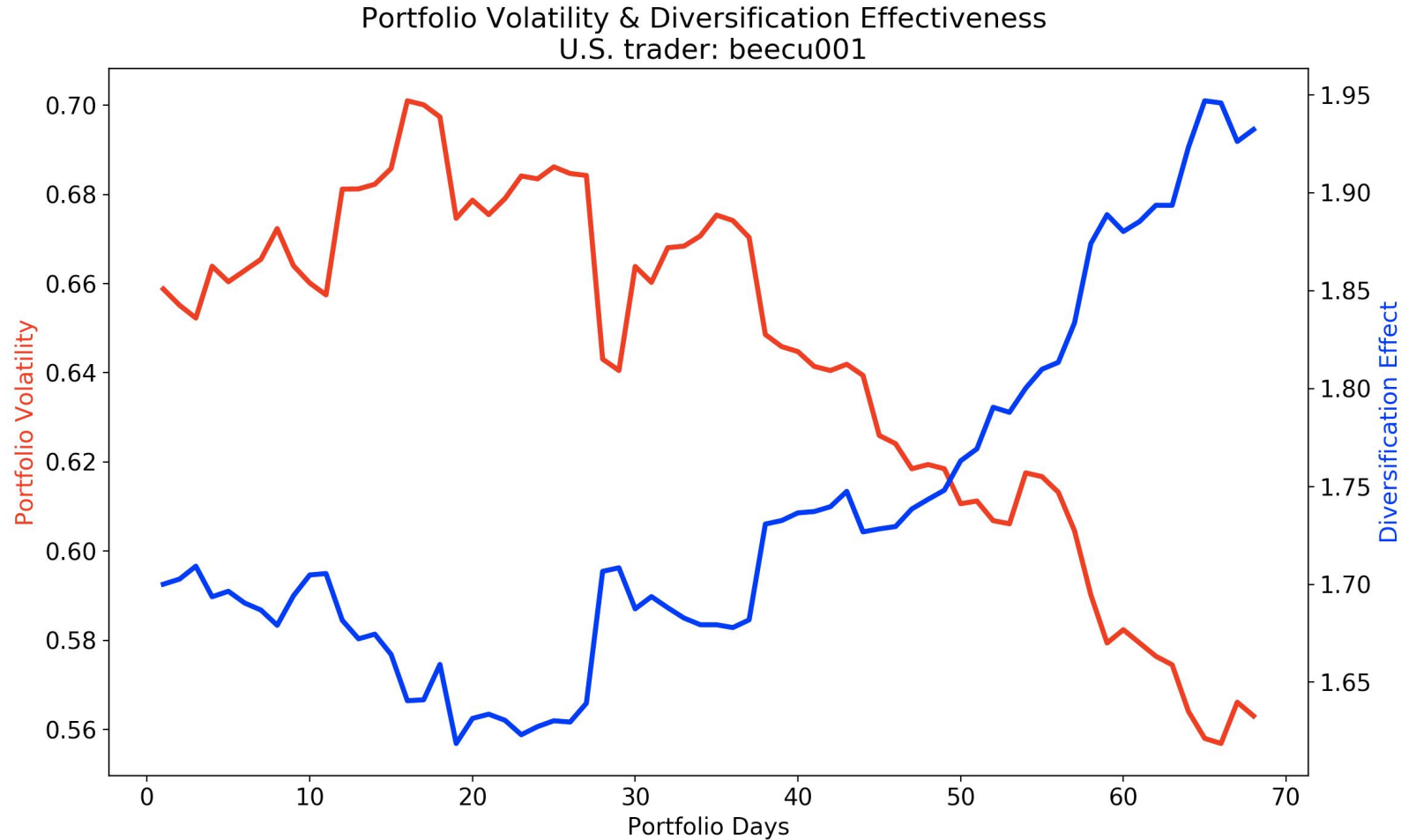


Fig. 4 Actual number of assets in portfolio



FDTScore: Hedging Effect Analysis – ‘beecu001’



Initially, when DE was lower ~1.7, portfolio volatility was higher ~0.7; then the trader’s diversifying effect increased in the latter half to 1.95, effectively reducing portfolio risk to 0.56.

Portfolio Strategy “beecu002”

HK Stock

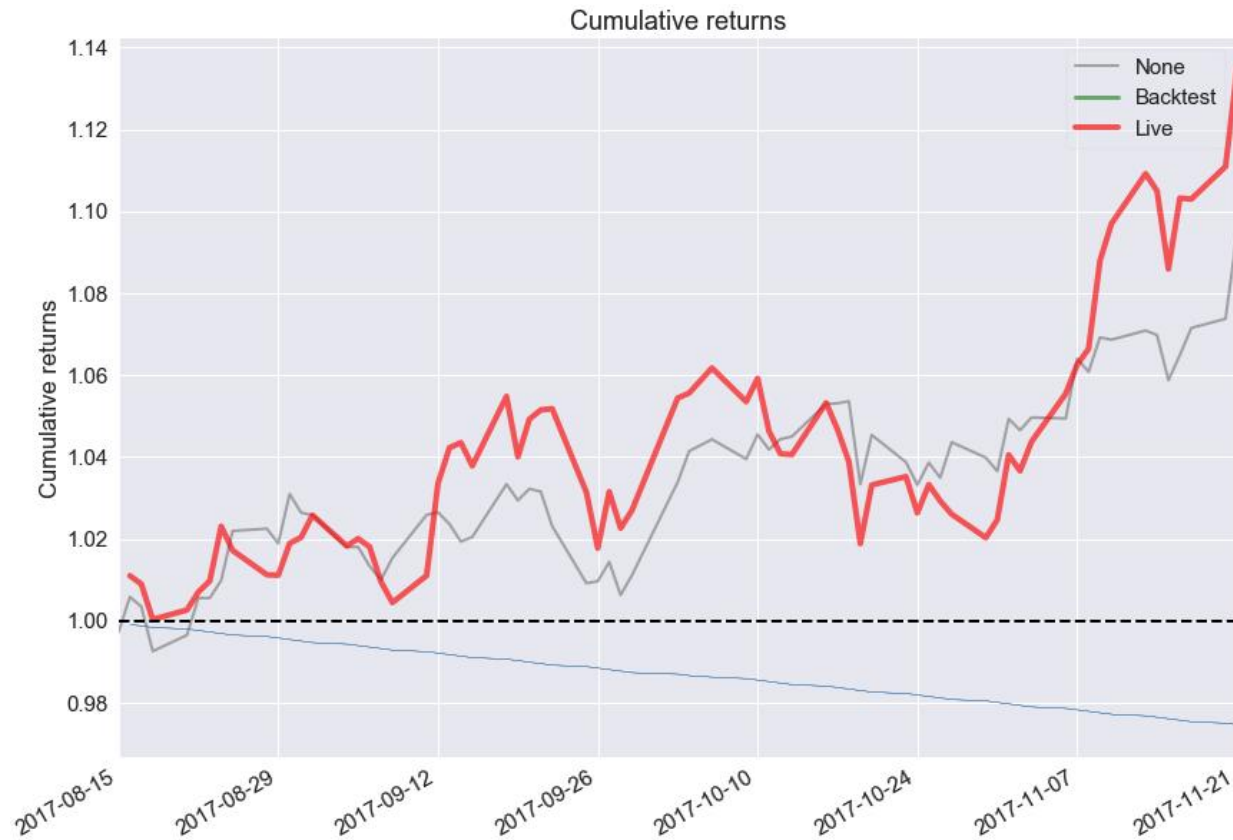


• Portfolio “beecu002”

- Started on 2017-08-14 (68 days)
- Market: HK Stock
- Benchmark: HS Index

- Alpha: 0.15
- Beta: 0.95
- Max Drawdown: -4.0%

- Annual Ret.: 58.4%
- Annual Volatility: 15.7%



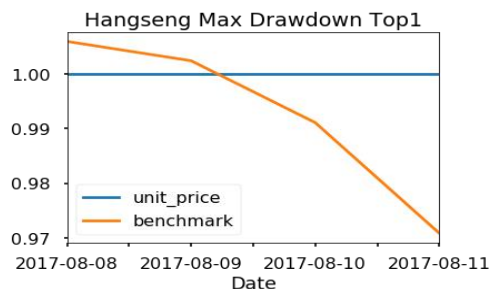
Portfolio Strategy “beecu001”

Blue: strategy

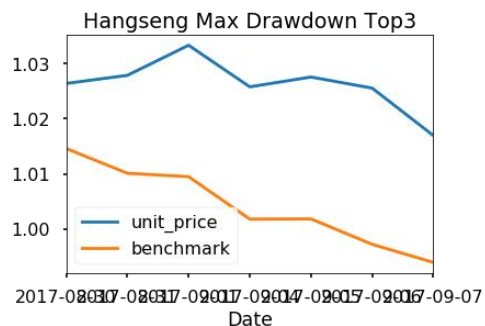
Orange: Benchmark

FDTAI

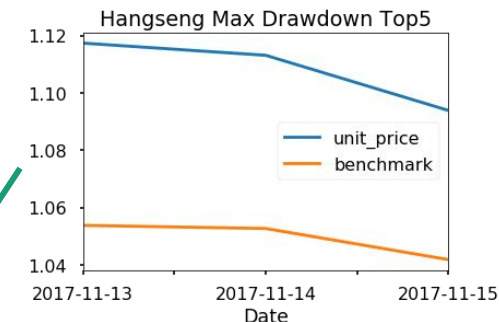
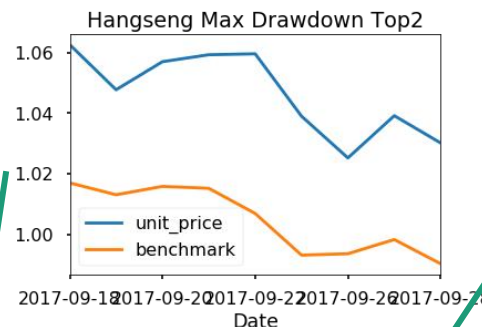
- Drawdown analysis



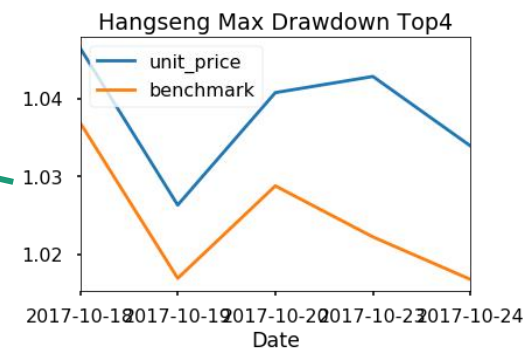
Top-1



Top-3



Top-5



Top-4



FDTScore: Diversity Analysis - "beecu002"

HK Stock

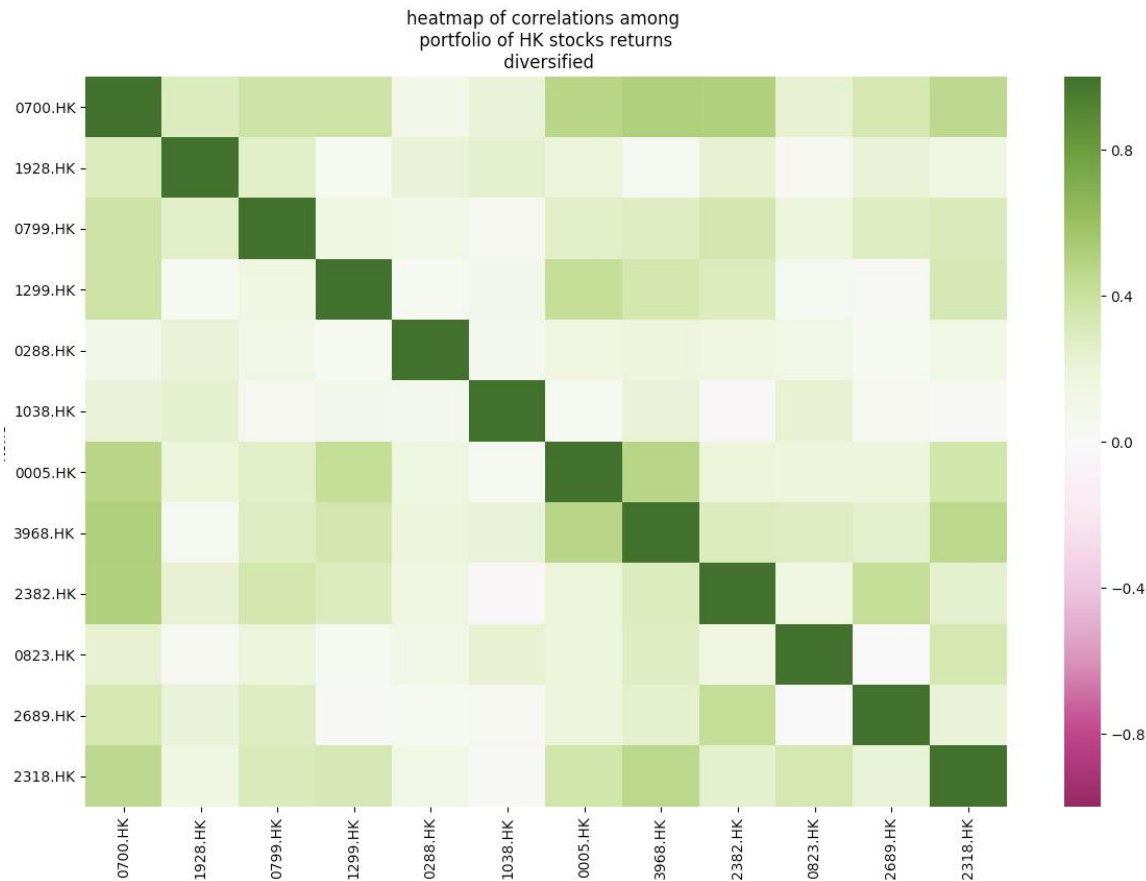
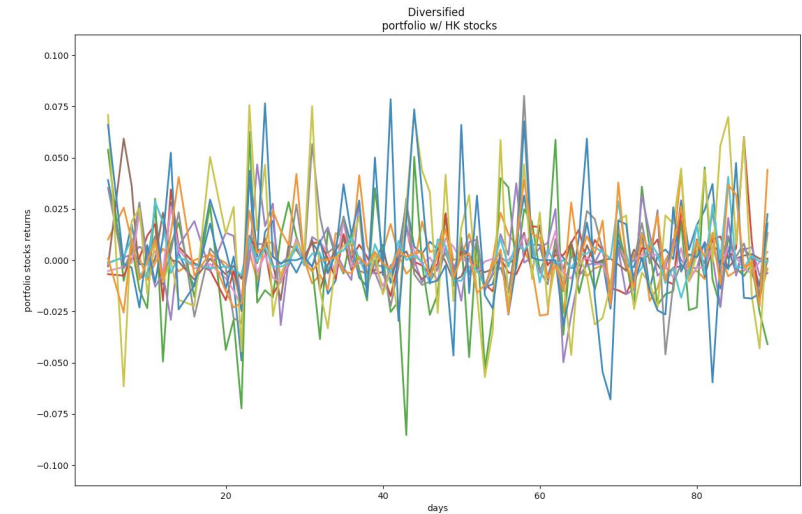


Fig. 1 Heatmap of correlations among portfolio assets' returns - Diversified (beecu002)

Portfolio Stocks Returns



Diversified Stocks

- Remarks
 - Both plot shows the trader's portfolio being not as diversified.

FDTScore: Soft-unique Analysis – ‘beecu002’,^{HK Stock}

Fig. 3 Number of assets explaining portfolio variance

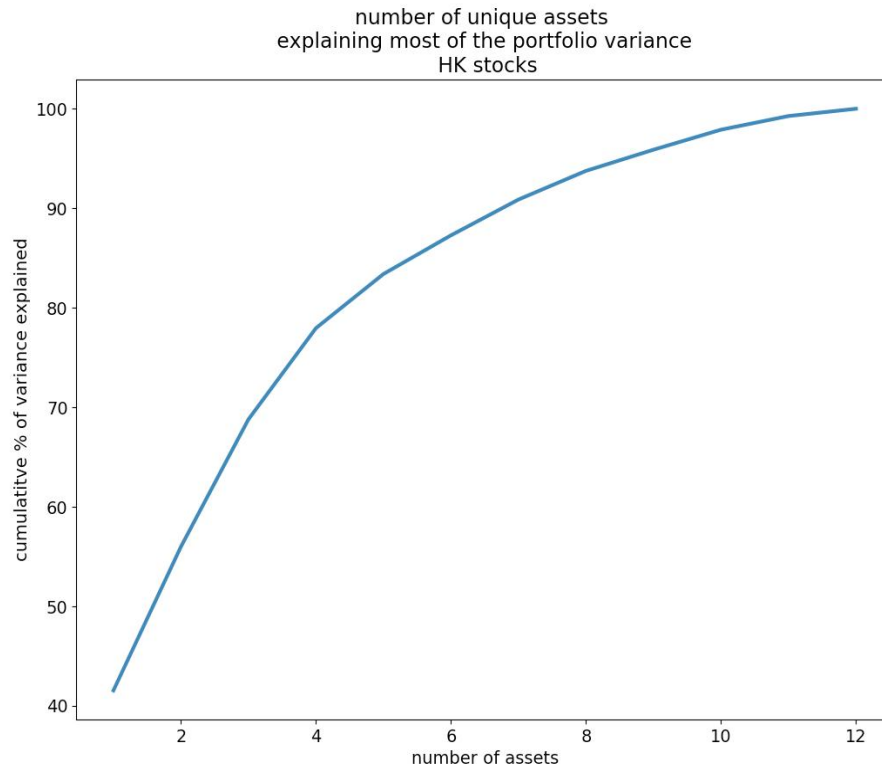
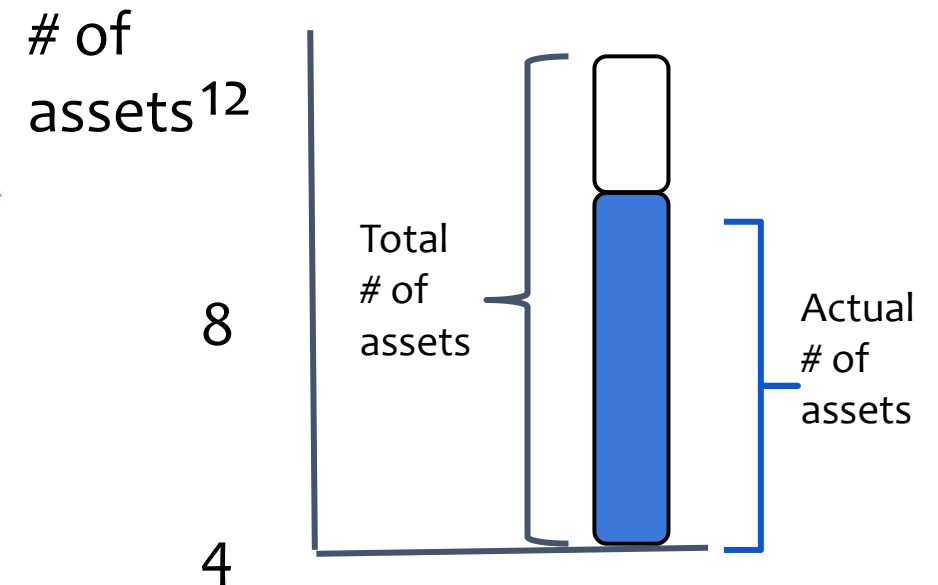
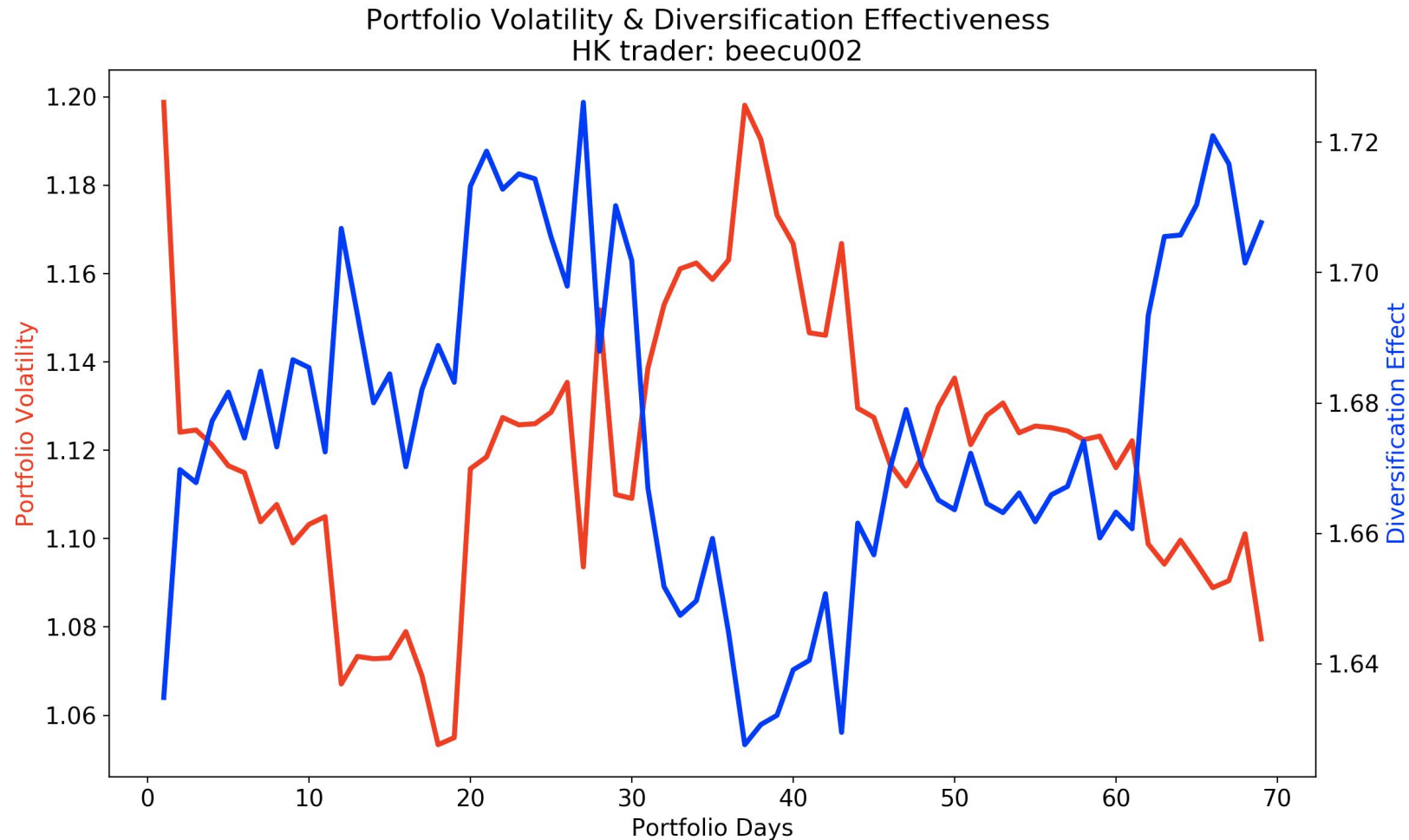


Fig. 4 Actual number of assets in portfolio



FDTScore: Hedging Effect Analysis – ‘beecu002’



DE in HK market first increased then decreased and finally increased again, hence, portfolio risk was low then high and finally lowered again. Portfolio volatility range from 1 to 1.2, while DE ranges from 1.6 to 1.72

Portfolio Strategy “beecu003”

China A-share



• Portfolio “beecu003”

- Started on 2017-09-15 (43 days)
- Market: China-A
- Benchmark: HS300
- Alpha: 0.10
- Beta: 1.30
- Max Drawdown: -2.3%
- Annual Ret.: 133.7%
- Annual Volatility: 13.6%

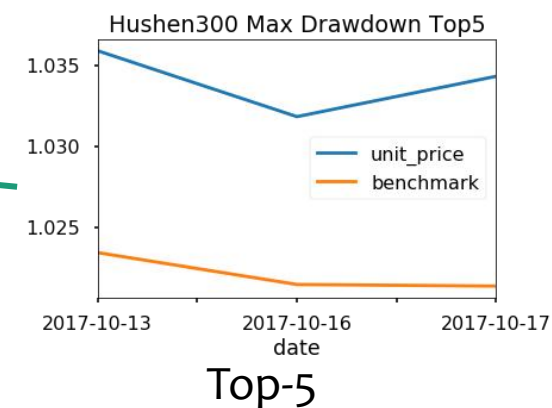
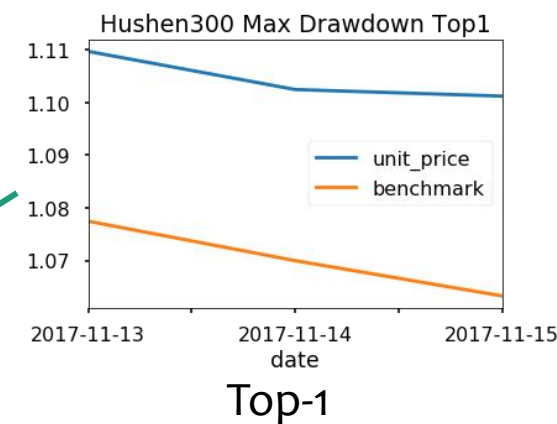
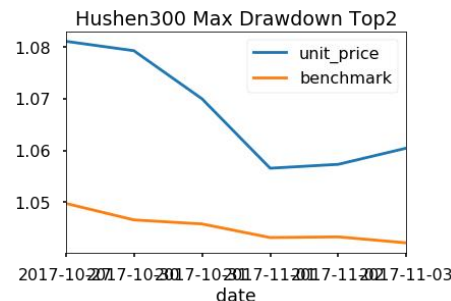
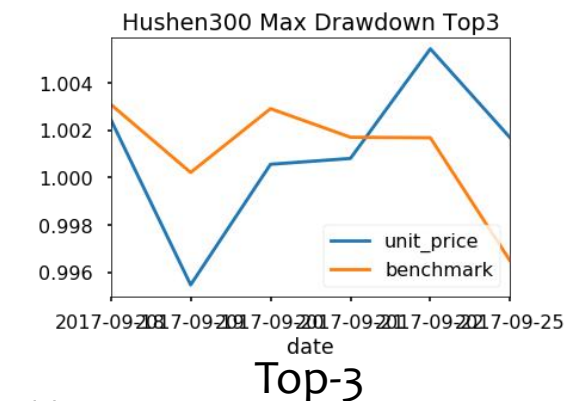
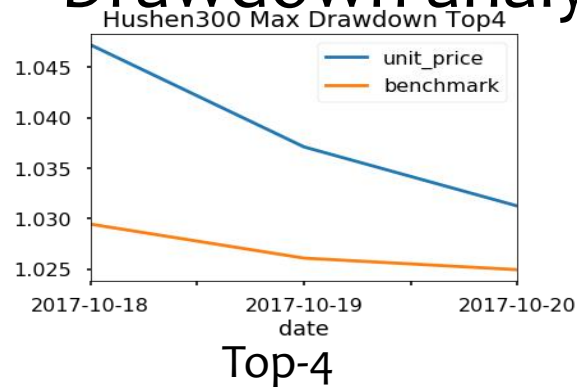
Portfolio Strategy “beecu001”

Blue: strategy

Orange: Benchmark

FDTAI

- Drawdown analysis



FDTScore: Diversity Analysis - "beecu003"

China A-share

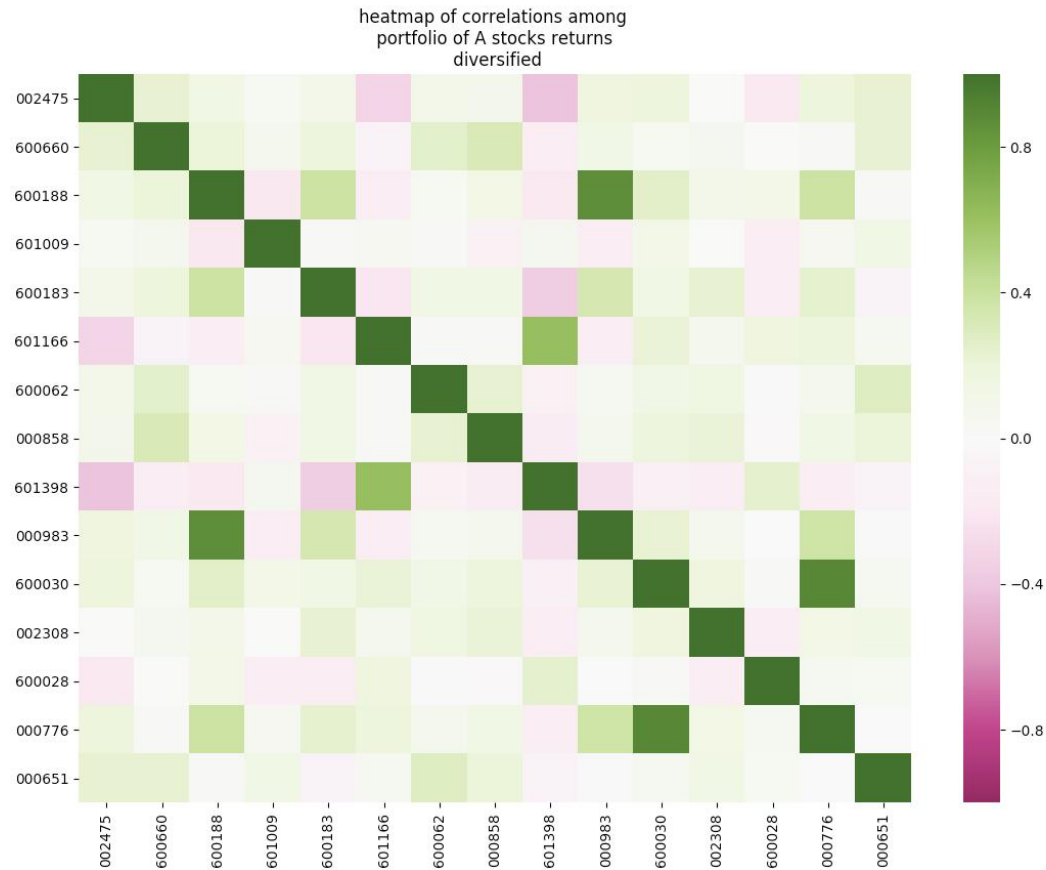
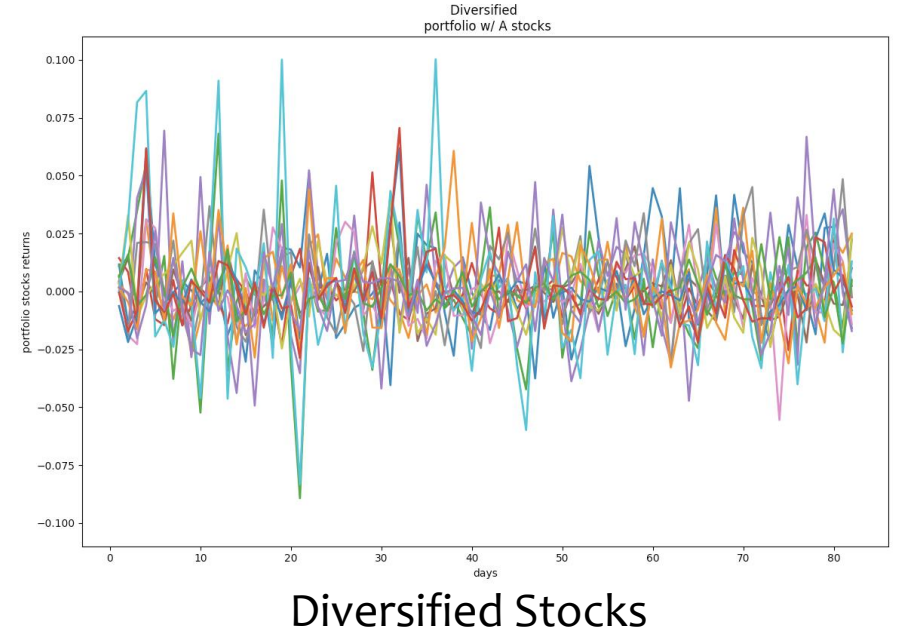


Fig. 1 Heatmap of correlations among portfolio assets' returns - Diversified (beecu002)

Portfolio Stocks Returns



• Remarks

- The plot shows the trader's portfolio being very diversified, with some degree of hedging.

FDTScore: Soft-unique Analysis – ‘beecu003’, China A-share

Fig. 3 Number of assets explaining portfolio variance

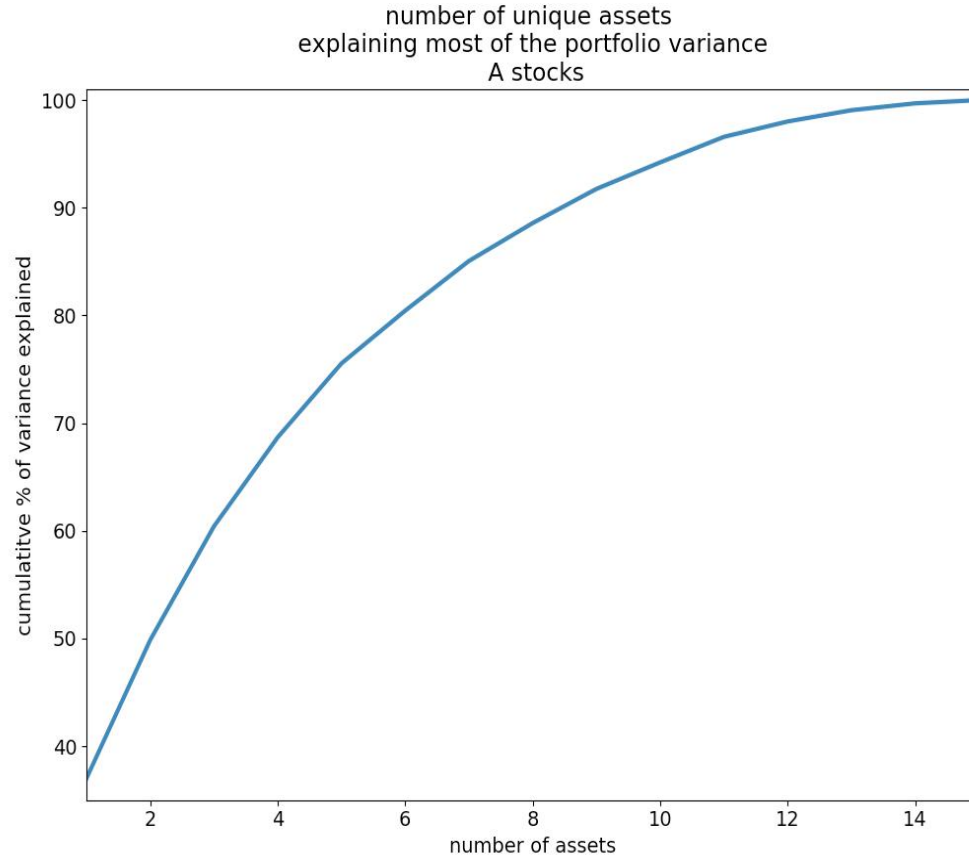
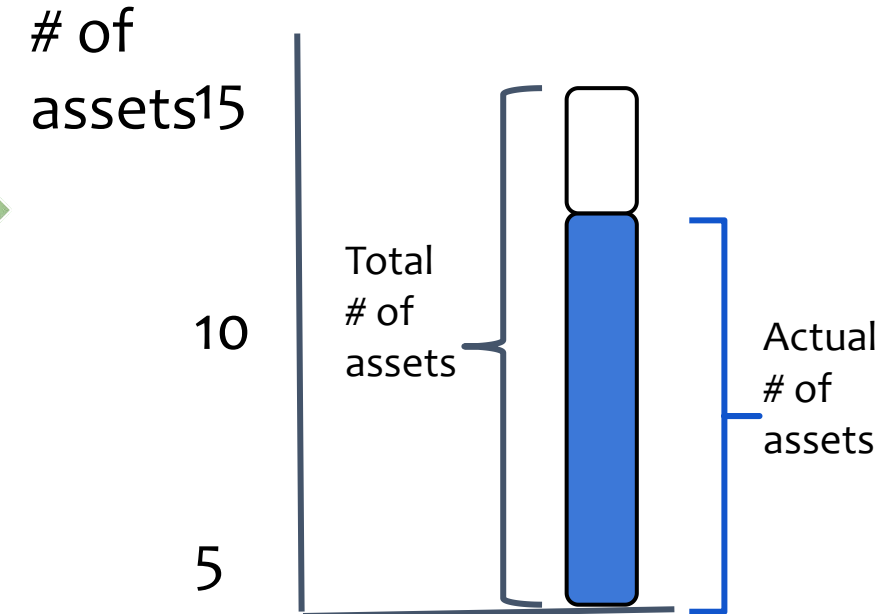


Fig. 4 Actual number of assets in portfolio



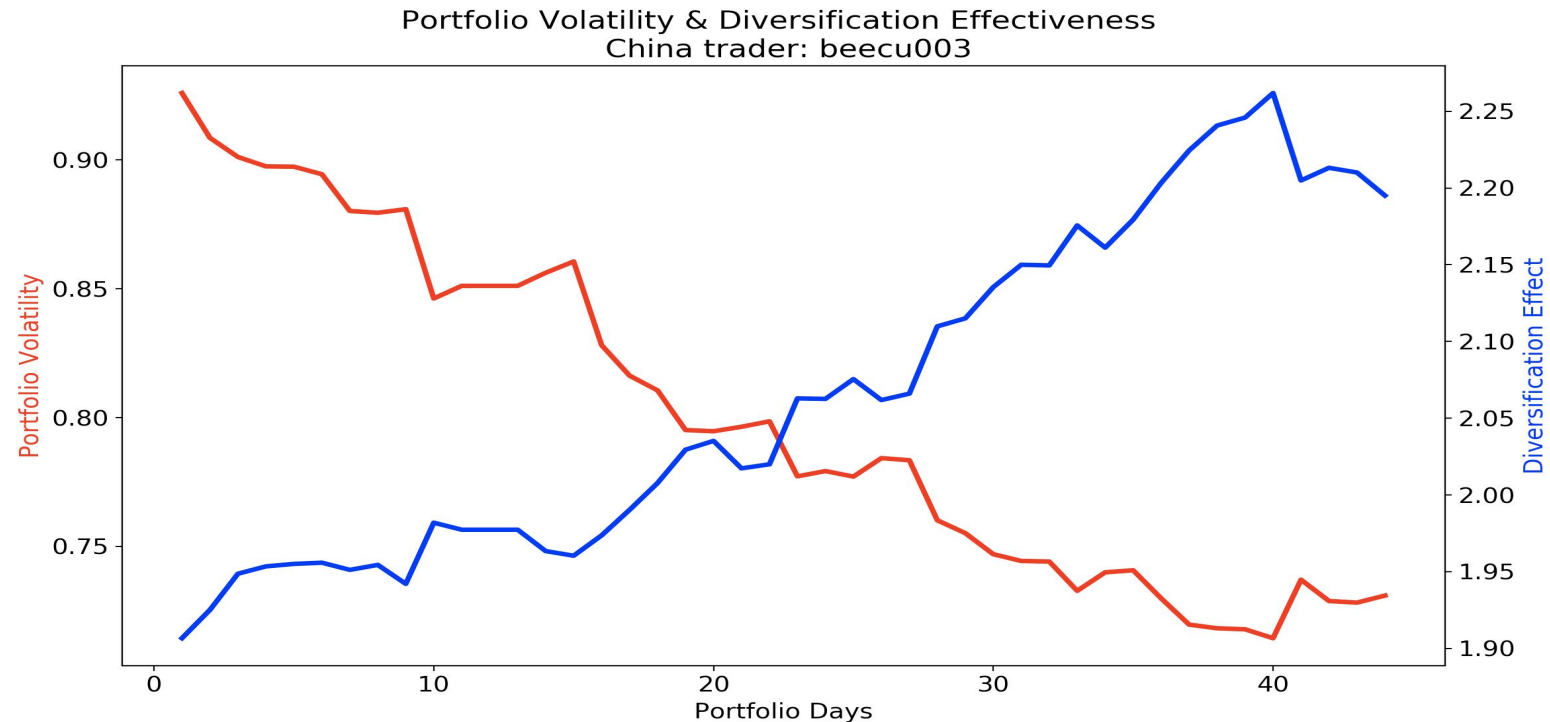
FDTScore: Hedging Effect Analysis – ‘beecu003’



The ultimate purpose of diversification is to reduce risk.

We can quantify the amount of risk decreased from diversification, using the feature diversification effectiveness.

$DE \in [1, \infty)$, where 1 indicates no diversification effect, larger number means better diversified



Initially, when DE was lower ~1.9, portfolio volatility was higher ~0.8 - 0.9; then the trader's diversifying effect continually increased all the way to >2.2, effectively reducing portfolio risk to < 0.7.

FDTScore: Diversity Analysis



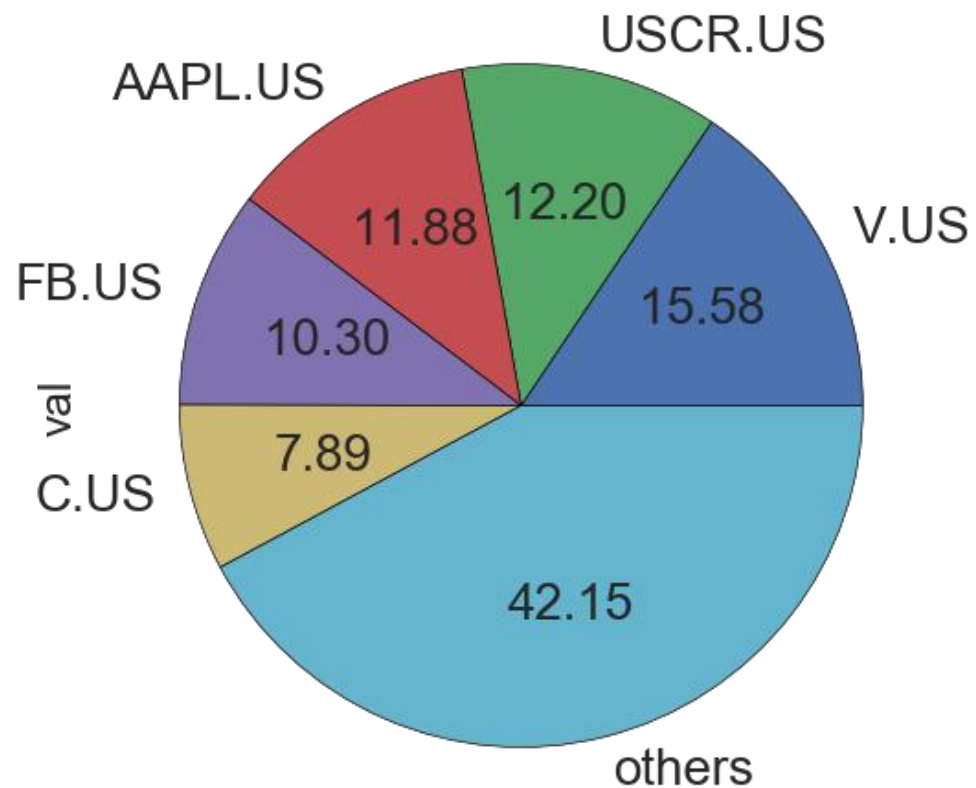
- How similar are your assets?
 - When you hold many assets, but many of your assets are similar, then you aren't really diversifying.

FDTScore: Diversity Analysis

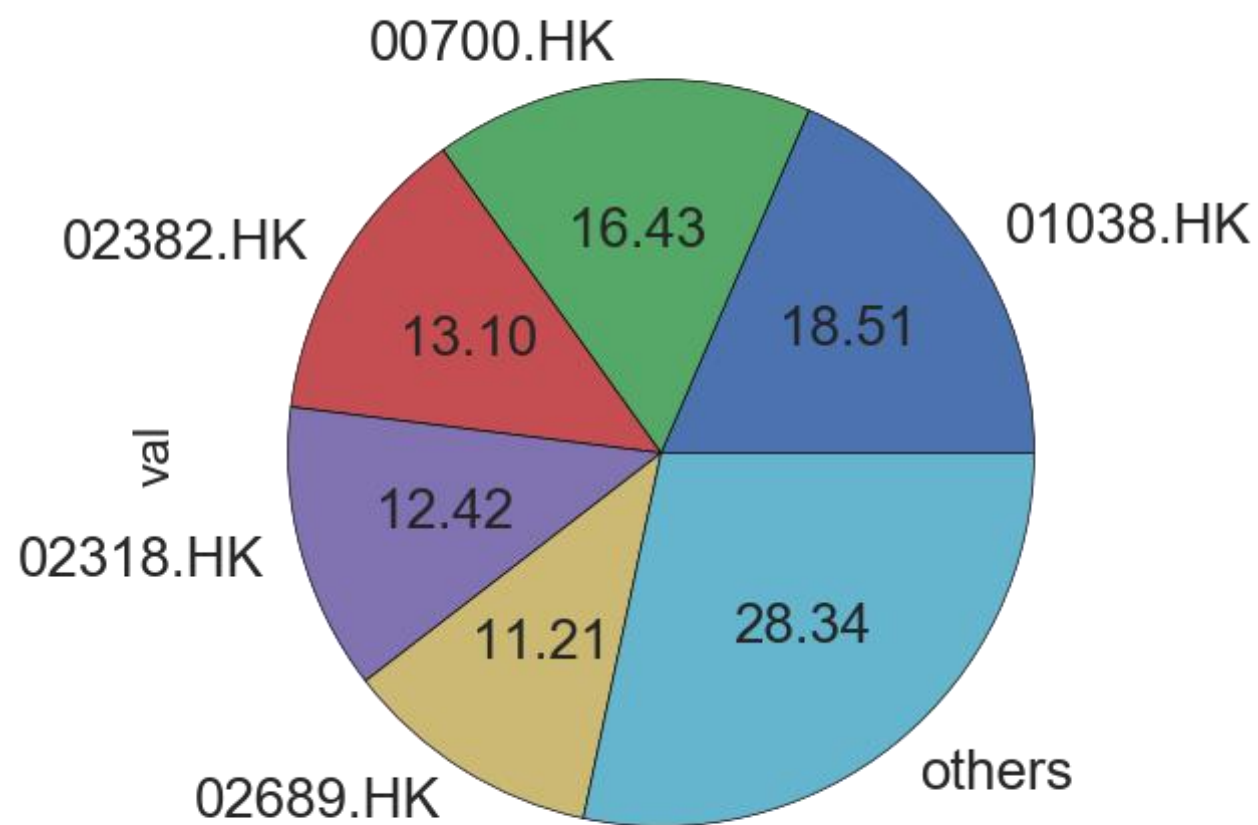
- Summary

1. The trader in A stocks ('beecu003') market is the most-diversified, even showing strategies of hedging.
2. The trader in US stocks ('beecu001') market is well-diversified, but not as good in A stocks market; there is less hedging.
3. The trader in HK stocks ('beecu002') market is the least-diversified, compared to the other two. As shown by the heatmap, similarities are higher among the assets.

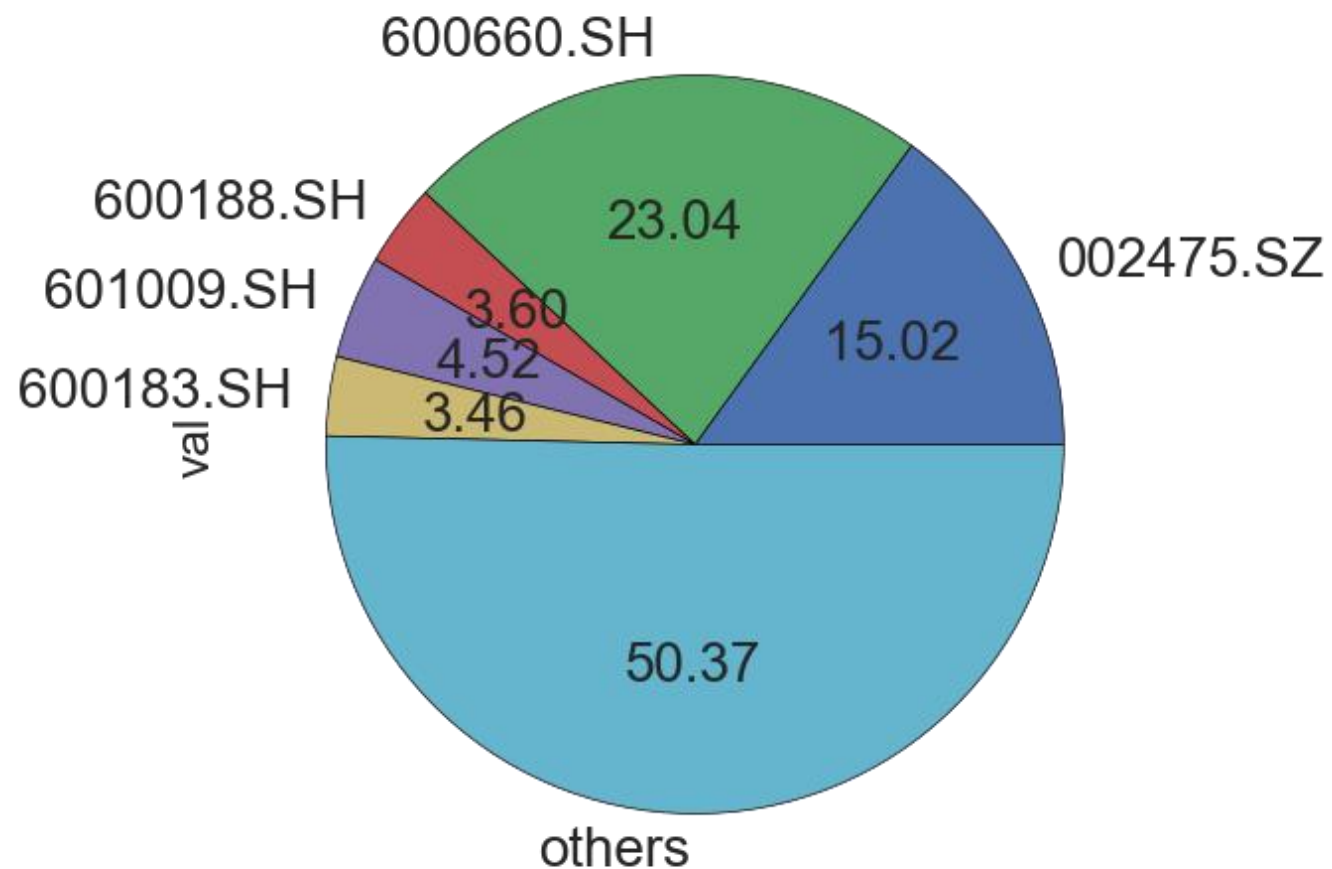
Heavy-position Assets – US Stock



Heavy-position Assets - HK Stock

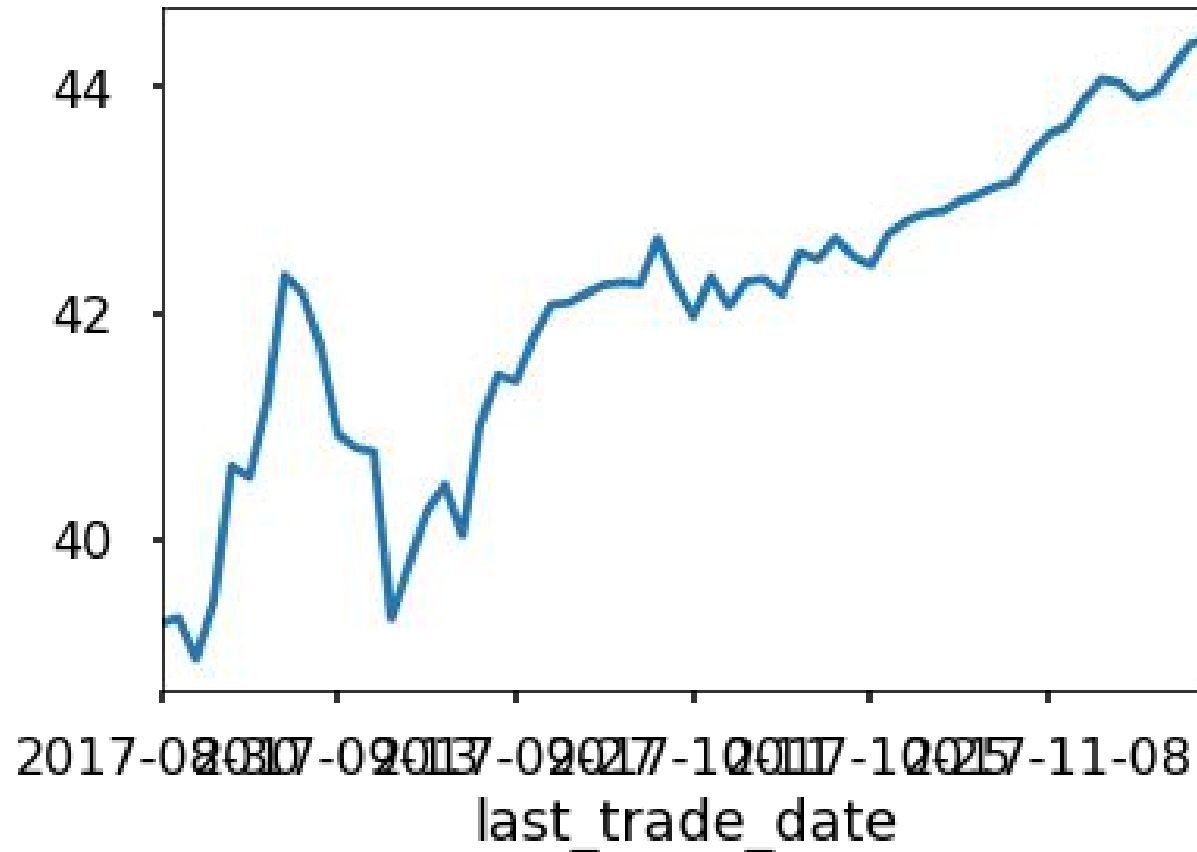


Heavy-position Assets – A share



Backup Slides

FDTScore Statistics



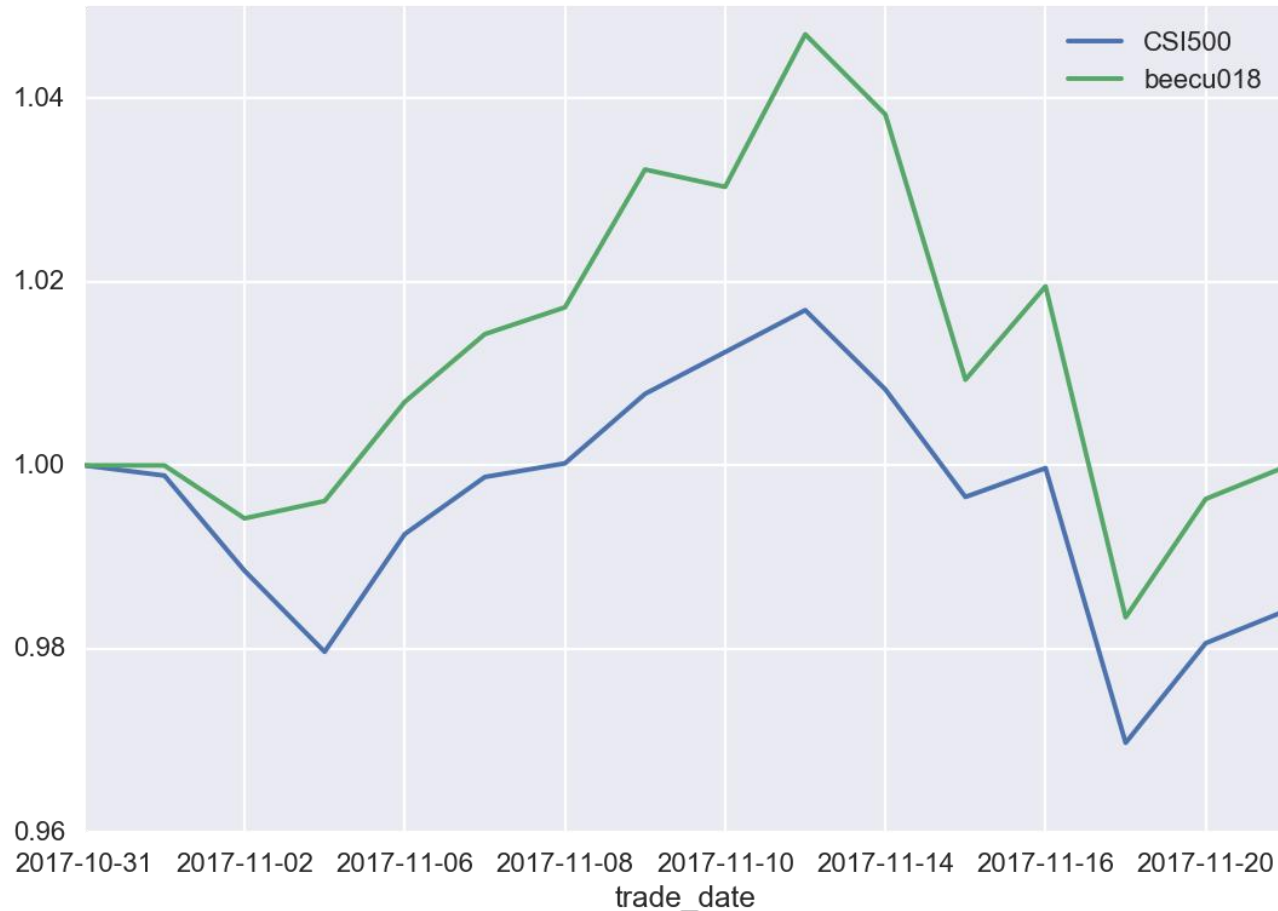
Portfolio Strategy “beecu008”



• Portfolio “beecu008”

- Started on 2017-10-31
- Market: China A
- Benchmark: HS300
- Alpha: -0.09
- Beta: 2.07
- Max Drawdown: -2.6%
- Annual Ret.: 281.8%
- Annual Volatility: 26.0%

Portfolio Strategy “beecu018”



• Portfolio “beecu018”

- Started on 2017-10-31
- Market: China A
- Benchmark: IC500
- Alpha: 0.39
- Beta: 0.98
- Max Drawdown: -6.1%
- Annual Ret.: -0.2%
- Annual Volatility: 22%